

SUMMARIES

Maxwell's Equations

Gauss's Law

$$\oiint_S \vec{E} \cdot d\vec{S} = \frac{Q_{\text{encl.}}}{\epsilon_0} = \frac{1}{\epsilon_0} \iiint_{\mathcal{V}} \rho \, d\tau = \frac{1}{\epsilon_0} \iint_S \sigma \, dS = \frac{1}{\epsilon_0} \int_L \lambda \, d\ell$$

Solenoidal Condition

$$\oiint_S \vec{B} \cdot d\vec{S} = 0$$

Faraday's Law (static)

$$\oint_C \vec{E} \cdot d\vec{\ell} = \epsilon = - \frac{d}{dt} \iint_S \vec{B} \cdot d\vec{S}$$

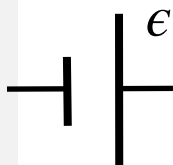
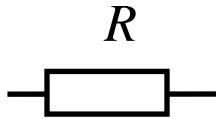
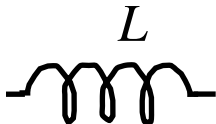
Ampère's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{encl.}} = \mu_0 \iint_S \vec{J} \cdot d\vec{S}$$

Maxwell introduced an extra term, i.e. displacement current, to Ampère's Law that turns E&M into electromagnetism!

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_S \vec{J} \cdot d\vec{S} + \mu_0 \epsilon_0 \frac{d}{dt} \iint_S \vec{E} \cdot d\vec{S}$$

Electrical elements

	battery	capacitor	resistor	inductor
symbol				
parameter	emf	capacitance	resistance	self-inductance
	ϵ	$C = \frac{Q}{V}$	$R = \frac{V}{I}$	$L = \frac{N\Phi_B}{I}$
voltage	ϵ	$\frac{Q}{C}$	RI	$L \frac{dI}{dt}$ drop
energy flow	provides	stores in $\vec{E}$	dissipates	stores in $\vec{B}$
energy store		$\frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$		$\frac{1}{2} LI^2$
power	VI in		RI^2 out	
in series	$\epsilon_1 + \epsilon_2$	$\left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1}$	$R_1 + R_2$	$L_1 + L_2$
in parallel	$\epsilon_1 = \epsilon_2$	$C_1 + C_2$	$\left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$	$\left(\frac{1}{L_1} + \frac{1}{L_2} \right)^{-1}$
impedance	ϵ	$\frac{1}{j\omega C}$	R	$j\omega L$
geometry		$C = \frac{\epsilon A}{d}$	$R = \frac{\rho \ell}{A}$	$L = \frac{\mu N^2 A}{\ell}$

Essential formulae

Coulomb Force

$$\vec{F}_{12} = \frac{q_1 q_2}{4\pi\epsilon_0} \frac{(\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|^3}$$

Electric Field

point charge: $\vec{E}(\vec{r}) = \lim_{q_0 \rightarrow 0} \frac{\vec{F}_0}{q_0}$ $\vec{F}_i = q_i \vec{E}(\vec{r}_i)$ $\vec{E}(\vec{r}) = \frac{q_i}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|^3}$

continuous charge distributions: $\vec{E}(\vec{r}) = \int \frac{\lambda(\vec{r}')}{4\pi\epsilon_0} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} d\ell'$

$$\vec{E}(\vec{r}) = \iint \frac{\sigma(\vec{r}')}{4\pi\epsilon_0} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dS' \quad \vec{E}(\vec{r}) = \iiint \frac{\rho(\vec{r}')}{4\pi\epsilon_0} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} d\tau'$$

dipole in external field: $\vec{p} = q\vec{d}$ $\vec{\tau} = \vec{p} \times \vec{E}$ $\mathcal{U}(\phi) = -\vec{p} \cdot \vec{E}$

field of an ideal dipole: $\vec{E} \approx \frac{1}{4\pi\epsilon_0 r^3} [3(\vec{p} \cdot \hat{r})\hat{r} - \vec{p}]$

Gauss's Law

$$\Phi_E = \iint_S \vec{E} \cdot d\vec{S} \quad \oiint_S \vec{E} \cdot d\vec{S} = \frac{Q_{\text{encl.}}}{\epsilon_0}$$

Electrostatic energy and potential

$$W_{a \rightarrow b} = \mathcal{U}_a - \mathcal{U}_b = \int_a^b \vec{F} \cdot d\vec{\ell} \quad V(\vec{r}) = \lim_{q_0 \rightarrow 0} \frac{\mathcal{U}_{10}}{q_0} \quad V_{ab} = V_a - V_b = \int_a^b \vec{E} \cdot d\vec{\ell}$$

point charge: $\mathcal{U}_{12} = \frac{q_1 q_2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|}$ $V(\vec{r}) = \frac{q}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{x} - \frac{\partial V}{\partial y} \hat{y} - \frac{\partial V}{\partial z} \hat{z} \equiv -\vec{\nabla} V$$

$$\mathcal{U} = \sum_{i=1}^n \frac{1}{2} q_i V_i \quad u = \frac{1}{2} \epsilon_0 E^2$$

Current

$$I = \frac{dQ}{dt} \quad \vec{J} = \frac{I}{A} \hat{z} = n q \vec{v}_d \quad \vec{E} = \rho \vec{J} R = \frac{\rho L}{A} \quad V = R I$$

Magnetic field

Lorentz force: $\vec{F} = q \left(\vec{E} + \vec{v} \times \vec{B} \right)$

Force on current wire: $\vec{F} = \int I d\vec{\ell} \times \vec{B}$

Solenoidal condition: $\oiint_S \vec{B} \cdot d\vec{S} = 0$

Magnetic dipole: $\vec{\tau} = \vec{\mu} \times \vec{B}$, $\vec{\mu} = I A \hat{n}$

Law of Biot-Savart

$$\vec{B}(\vec{r}, t) = \frac{\mu_0}{4\pi} q \frac{\vec{v}(t) \times (\vec{r} - \vec{r}'(t))}{|\vec{r} - \vec{r}'(t)|^3} \quad \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int I \frac{d\vec{r}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

Ampère's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{encl.}}$$

Faraday's Law

$$\epsilon = - \frac{d\Phi_B}{dt}$$

motional emf: $\epsilon = \oint_{\text{loop}} (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$

stationary loop: $\epsilon = \oint_C \vec{E} \cdot d\vec{\ell}$

$$u = \frac{\mathcal{U}}{\mathcal{V}} = \frac{B^2}{2\mu_0}$$

Materials

$$\epsilon_r = \frac{C}{C_0} \quad \vec{B} = \mu_r \vec{B}_0$$